

$f(x)$ - graphs height

$f'(x)$ - slope (positive rising, zero flat, negative falling)
 $\rightarrow$ The first derivative

Convex Slopes never decreases while x increases (falls slower)
 Concave ... increases ... increases (rises slower)

[Convexity describes how the slope changes,
 not whether the function itself rises or falls.]

Throughout the interval

$$f''(x) \geq 0$$

$$f''(x) \leq 0$$

Meaning

Slope never decreases

Slope never increases

Classification

Convex

Concave

$$f(x) = x^2 \rightarrow f'(x) = 2x \rightarrow f''(x) = 2 \quad \text{Convex}$$

- Check the whole interval; one point does not establish convexity over the whole domain.

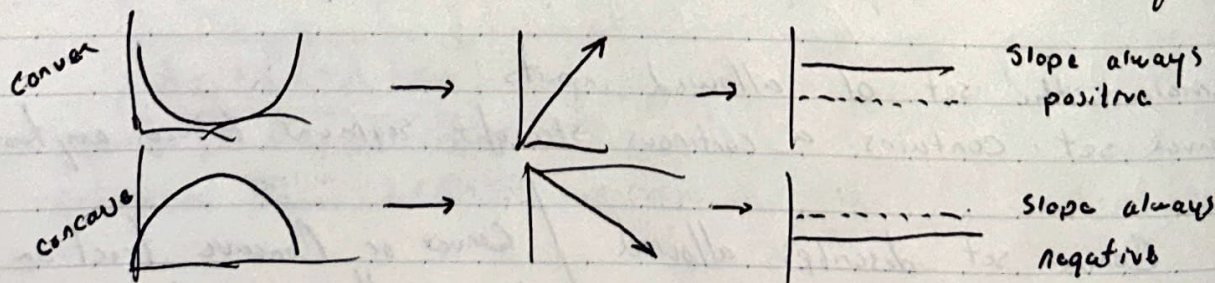
~~The slopes direction of change must both throughout the interval~~
~~negative answers~~ Convexity means being convex

Classify this function over all real numbers

$$f(x) = x^2 - 4x + 3 \rightarrow f'(x) = 2x - 4 \rightarrow f''(x) = 2 > 0$$

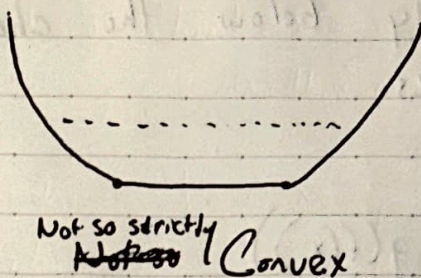
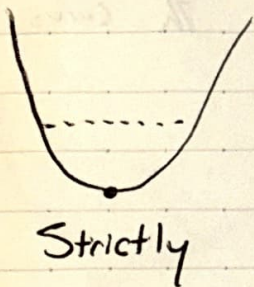
$\rightarrow$ ~~Concave~~ Convex

Any ~~convex~~ Chord has a height higher than the curve height



At inflection points concavity changes.

Strictly Convex requires chord to stay strictly below curve
 → can have at most 1 minimum point.



Calculating Derivatives

Sublevel Sets

$A_a = \{x: f(x) \leq a\}$ "All ~~inputs~~ whose output is at most a ."

example: $f(x) = x^2$ and height limit = 4

$$A_4 = \{x: x^2 \leq 4\} = [-2, 2]$$

Picture drawing a horizontal line at height 4, then collecting the inputs where the curve is at or below it. Those inputs form an interval with no gaps - a convex set.

[every sublevel set of a convex function is convex]

if the cost function is convex blending two affordable choices stays affordable. $f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda)f(x_2) \leq 100$

blended choice $\leq$ blended endpoint cost $\leq$ budget

Concave superlevel set
 $\{x: f(x) \geq a\}$

profit of blend $\geq$ average endpoint profits ≥ 100

[Convex sublevel sets do not prove itself is convex.]

2 plans of at least \$100

Convex vs. Strictly convex $\rightarrow$ Strict Chord height cannot equal actual curve height at any point.

- Strictly below the chord is the curve at all times.

Composition

$$h(x) = g(f(x))$$

if convex (convex) then convex

$\hookrightarrow$ & non-decreasing (large inputs never cause smaller values)

Chain rule

$$h'(x) = g'(f(x)) f'(x)$$

such $(3x+1)^2 \rightarrow 2(3x+1) \cdot (3) = 6(3x+1)$

for $(2x+5)^2 \rightarrow 2(2x+5) \cdot 2 = 4(2x+5) = 8x+20$

Tangent lines

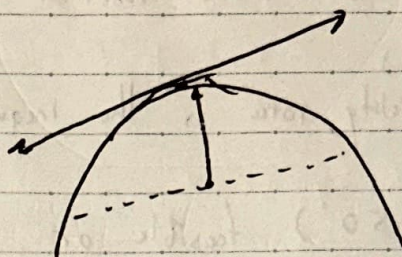
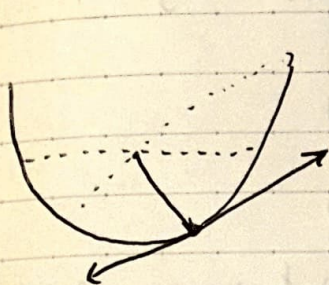
$$T(x) = f(a) + f'(a) \cdot (x-a)$$

such as $f(x) = x^2$ $a=1$

$$\hookrightarrow T(x) = 1 + 2(x-1) = 2x-1$$

at $x=2$ tangent predicts height at 3 when curves actual is 4.

	Chord	Tangent
Convex is ...	Below / overestimate	Above / underestimate
Concave is ...	Above	Below



Slope at a is zero $f'(a) = 0$

Then inequality becomes $f(x) \geq f(a)$

every other value is at least as high as a
so a is a global minimum (if convex).

9/23 Class

Feasibility and Constraints

At a specific feasible point $\bar{x}$, an equality constraint $g_i(x) \geq 0$ is called active (or binding) if $g_i(\bar{x}) = 0$ and inactive (or nonbinding) if $g_i(\bar{x}) > 0$.

Active set at a feasible point is the set of all constraints that is active past a point

Boundary - all feasible points set...

Interior points -